

MTH210 – Statistical Computing Project

NAME : Jani Ravi Kailash

ROLL NO : 240486

File Used: fort.11

1. Introduction

In this project, I am given a dataset generated from the following probability density function:

$$\begin{aligned} &[\\ f(x; \alpha, \lambda) &= \alpha \lambda e^{-\lambda x} (1 - e^{-\lambda x})^{\alpha - 1}, \quad x > 0 \\ &] \end{aligned}$$

where $\alpha > 0$ and $\lambda > 0$.

The objective is to estimate the unknown parameters α and λ using the **Newton-Raphson method**, and then construct **95% confidence intervals** using both parametric and non-parametric bootstrap methods.

2. Methodology

2.1 Maximum Likelihood Estimation

The likelihood function was constructed from the given PDF and converted into a log-likelihood form for easier computation.

The Newton-Raphson method was applied using:

- First derivatives (score functions)
- Second derivatives (Hessian matrix)

The update rule used was:

$$\theta_{\text{new}} = \theta - H^{-1} U$$

where U is the gradient vector and H is the Hessian matrix.

2.2 Initial Guess

The initial values used were:

$$(\alpha, \lambda) = (1, 1)$$

2.3 Stopping Criterion

The algorithm was terminated when:

$$\max(|\theta_{\text{new}} - \theta|) < 10^{-6}$$

2.4 Number of Iterations

The Newton-Raphson method converged in: **6** iterations.

3. Bootstrap Methods

3.1 Parametric Bootstrap

In this method, new samples were generated using the estimated values of α and λ . The inverse transform method was used for simulation, and parameters were re-estimated for each generated sample.

3.2 Non-Parametric Bootstrap

In this case, samples were drawn with replacement from the original dataset. The parameters α and λ were estimated for each resampled dataset.

3.3 Number of Replications

The number of bootstrap replications used was:

$$B = 1000$$

4. Results

4.1 MLE Estimates

$$\hat{\alpha} = 2.0361$$

$$\hat{\lambda} = 0.906$$

4.2 Parametric Bootstrap 95% Confidence Intervals

$$\alpha: (1.2971, 4.8004)$$

$$\lambda: (0.6353, 1.5228)$$

4.3 Non-Parametric Bootstrap 95% Confidence Intervals

$$\alpha: (1.3819, 4.0621)$$

$$\lambda: (0.5836, 1.6993)$$

5. Conclusion

The parameters α and λ were successfully estimated using the Newton-Raphson method. The bootstrap methods provided reliable confidence intervals for both parameters.

Both parametric and non-parametric approaches gave consistent results, indicating stability of the estimation procedure.

6. Remarks

- The Newton-Raphson method converged efficiently with the chosen initial values
 - Bootstrap methods helped in assessing the variability of the estimates
 - The entire implementation was carried out in R and verified for correctness
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